MARKETSCOPE: GEOSPATIAL ANALYSIS FOR OPTIMAL SMALL-BUSINESS PLACEMENT WITH INTEGRATED REPORTING SYSTEM

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CHAPTER 1
INTRODUCTION

Background of Project

Entrepreneurs worldwide often face the dilemma of choosing an optimal business location, frequently relying on "gut feel" or purely observational methods rather than empirical data. This lack of data-driven decision-making consistently leads to a high business failure rate. Without a tool to analyze market gaps, small business owners enter markets "blind," risking their capital on locations already over-populated with similar services. To mitigate these risks, the integration of Geographic Information Systems (GIS) combined with multi-criteria decision-making models has emerged as a crucial strategic asset. Research by Baviera-Puig et al. demonstrates that mapping "geodemand" and "geocompetition" through GIS and analytical hierarchy processes significantly improves retail site selection and mitigates financial risks associated with poor placement [1]. Similarly, Ertuğrul and Erbaş emphasize that leveraging spatial data through GIS prevents the cannibalization effect and identifies underserved zones, enabling retail businesses to establish a presence where market demand aligns with balanced competition [2] .

Within the archipelagic economy of the Philippines, micro, small, and medium enterprises (MSMEs) are critical drivers of growth, yet they remain highly vulnerable to operational inefficiencies and poor strategic planning. A major hurdle is that small businesses generally cannot afford expensive corporate consultancy or comprehensive data research to guide their site selection. A study analyzing Philippine micro-enterprises highlights that MSMEs often rely on simple, non-sequential steps and informal information gathering when making strategic decisions, which frequently contributes to unsatisfactory firm performance and high failure rates [3]. Furthermore, research evaluating MSE management practices in the Philippines indicates that formalized operational functions, particularly in strategic planning and market analysis, play a vital role in business success, yet these are often underutilized due to resource constraints [4]. Solving the problem of poor placement goes beyond individual business success; it is a necessary step to help local governments manage urban growth and ensure a more balanced distribution of services across communities.

In the specific context of Davao del Norte, Panabo serves as the primary economic hub for Panabo City, making it the most critical environment for small business placement and competition analysis. The recurring issue in this high-activity zone is unsustainable market saturation, where high-potential zones remain underserved while others become heavily congested. Investigations into MSMEs within the Davao Region reveal that most micro and small establishments are managed by a single person who simultaneously handles all administrative functions, leaving them without the time or financial capacity to conduct adequate market research or establish a focused target market [5]. Additionally, these local enterprises are highly susceptible to environmental turbulence, making data-backed resilience strategies essential for survival [6]. To address these localized challenges, MarketScope is developed as an Intelligent Decision Support System (IDSS) that utilizes geospatial analysis to provide data-driven placement recommendations. By integrating a spatial evaluation engine and ensuring regulatory compliance through City Land Use Plan (CLUP) data, MarketScope levels the playing field by transforming complex geospatial and demographic data into actionable insights for the average entrepreneur. Furthermore, it utilizes Generative AI to translate complex spatial variables into accessible feasibility reports.

Crucially, the development of MarketScope is deeply anchored in Davao del Norte State College's (DNSC) institutional sustainability goals, which are strictly aligned with the United Nations 2030 Agenda for Sustainable Development. By democratizing access to complex geospatial data and market analytics, the system directly supports DNSC’s initiatives for SDG 8 (Decent Work and Economic Growth) and SDG 9 (Industry, Innovation, and Infrastructure) by fostering technological innovation and economic resilience among local micro, small, and medium enterprises (MSMEs). Furthermore, the project champions DNSC’s commitment to SDG 11 (Sustainable Cities and Communities) , a core focus of the institution's sustainable thinking and environmental stewardship campaigns. By ensuring regulatory compliance through City Land Use Plan (CLUP) data and utilizing AI-driven scoring, MarketScope transforms complex spatial variables into actionable insights that prevent unsustainable commercial congestion, securing entrepreneurial investments while promoting balanced, well-planned urban development.

The study is centered in Panabo City, Davao del Norte, specifically focusing on the high-activity zones of Panabo Proper. This area was selected because it serves as the primary economic hub for the city, making it the most critical environment for small business placement and competition analysis. The scope of the project is exclusively focused on these geographical boundaries, where the primary data clustering will occur to provide targeted insights for the local economy.

Within the high-activity zones of Panabo Proper, the proponents aim to address a sequence of interconnected issues that severely impact the viability of local micro, small, and medium enterprises (MSMEs). Primarily, new entrepreneurs often rely on "gut feel" or purely observational methods when choosing a business location, which frequently contributes to high business failure rates. Because these entrepreneurs lack data-driven decision-making tools, they tend to open new shops in immediate proximity to existing competitors, leading to unsustainable market saturation while high-potential zones remain unidentified. Beyond market competition, business owners are often unaware of environmental vulnerabilities, such as flood-prone areas, which can lead to significant inventory loss and operational downtime. The proponents seek to resolve the difficulty local entrepreneurs face in accessing and analyzing complex geospatial and risk data, which currently leaves them to enter hazardous or saturated markets "blind" and risk their capital.

Furthermore, prospective business owners face significant challenges regarding regulatory alignment, environmental resilience, and strategic planning.

There is a limited awareness of the City Land Use Plan (CLUP) and its associated zoning regulations, risking non-compliance and legal complications during storefront selection. Additionally, small businesses cannot afford expensive corporate consultancy for deep market research or formal feasibility studies. To overcome the lack of localized demographic census data, there is a critical need for a system that leverages highly accessible infrastructure proxies such as building density, road network classifications, and proximity to traffic generators to accurately gauge market demand and operational safety. Recognizing that the majority of local micro-entrepreneurs access digital tools via smartphones, establishing a mobile-responsive platform is critical for true technological accessibility.

To address these localized challenges, the project team from the Institute of Computing developed MarketScope, a mobile-optimized, web-based Intelligent Decision Support System (IDSS). The system utilizes geospatial analysis to provide data-driven recommendations for optimal small-business placement in Panabo City. It equips aspiring entrepreneurs with a mapping interface that visualizes existing business density and environmental hazard overlays. Specifically, the system integrates a visual flood zone layer, indicated by distinct red marks on the map, which is directly bound to the mathematical calculation of the business's suitability score, ensuring disaster resilience is strictly factored into the analysis. Furthermore, to enhance entrepreneurial strategy, the system introduces a proactive business trend and suggestion module. Instead of the user strictly searching for a location for a specific business, this feature allows "the business to find the user" by analyzing spatial data and recommending the most optimal MSME categories for a given area. Ultimately, MarketScope levels the playing field for MSMEs by transforming complex spatial variables, infrastructure proxies, and hazard data into accessible, actionable insights.

At its core, MarketScope is closely aligned with Davao del Norte State College's (DNSC) Research, Development, and Extension (RDE) Agenda 2022–2026. The project supports the institution's Technology and Innovation Thrust by delivering a practical geospatial solution that reflects applied, real-world research focused on the business environment of Panabo City. By making complex geospatial data and market insights more accessible, MarketScope helps local micro, small, and medium enterprises (MSMEs) make better decisions and improve their chances of long-term sustainability [11].

The system highlights DNSC's commitment to developing technology-based solutions that directly benefit the local community. Through the integration of Panabo City's City Land Use Plan (CLUP) data and the use of AI-driven scoring, MarketScope simplifies complex spatial information into clear and actionable insights. This helps prevent overcrowding of businesses in certain areas, protects entrepreneurial investments, and supports more balanced and well-planned urban development within the city's key commercial zones. Overall, the system serves as a practical example of how technology can support smarter, more effective local governance in line with the institution's RDE priorities.

Objectives of the Study

The primary goal of this study is to develop MarketScope, a decision support system that utilizes geospatial analysis to provide data-driven recommendations for optimal small-business placement in Panabo City.

The specific objectives are as follows:

// 1. Train and test TF-IDF, Word2Vec, and SBERT algorithms to determine the most accurate and efficient models for a content-based recommender system.

1. To design a web-based mapping interface that visualizes existing business establishments, local infrastructure, and an environmental hazard overlay.
2. To develop a spatial evaluation engine tailored for thirteen (13) MSME categories that allows users to assess an enterprise, utilizing building density, road classifications, and proximity to traffic generators.
3. To integrate the Panabo City Land Use Plan into the digital map, automatically cross-referencing user-selected locations against local commercial and residential zoning ordinances.

4. To develop a reporting module powered by generative artificial intelligence that translates the raw map, zoning, and proxy data into clear, context-aware, and downloadable business feasibility reports for prospective entrepreneurs.
5. To develop a proactive business trend and suggestion module that identifies optimal business types for specific areas and user's preferences.

Significance of the Study

The study introduces a data-driven, geospatial approach to small business placement, transforming traditional "gut-feel" decision-making into strategic, intelligent market analysis. It enhances entrepreneurial success rates, promotes balanced urban development, and levels the playing field for local MSMEs, while contributing to the growing field of localized business intelligence. The following are the study's main beneficiaries:

- **Aspiring Entrepreneurs and MSME Owners.** By utilizing this system, prospective business owners can minimize financial risks associated with poor location choices, easily identify market gaps, and access automated feasibility reports without the need for expensive corporate consultancy. This will increase operational confidence and long-term business survivability.
- **Local Government Unit.** The system will assist the local government in managing balanced urban growth by preventing unsustainable market saturation in concentrated areas. It also ensures that proposed business locations strictly align with the City Land Use Plan (CLUP) and local zoning regulations.
- **Local Consumers and Residents.** By guiding businesses toward underserved, high-potential areas, residents will gain more convenient access to local goods and services. This reduces the need to travel to heavily congested commercial centers and improves overall community resource distribution.

- **Future Researchers and Developers.** The foundation for future research on integrating geospatial mapping and AI-driven decision support systems in local economic development is laid by this study. It has the potential to spur innovations that will benefit not only Panabo City but also a broader range of municipalities seeking to optimize their local business landscapes.

Scope and Limitation

The scope of this project is strictly confined to the geographical boundaries of Panabo City, Davao del Norte, with the primary concentration of spatial analysis focused on the high-activity urban and commercial zones of Panabo Proper. The system is tailored specifically to assist aspiring entrepreneurs and owners of micro, small, and medium enterprises (MSMEs) who require data-driven guidance during the storefront selection process to ensure both commercial viability and physical disaster resilience. To ensure system accuracy and reliability, the prototype's spatial evaluation engine is explicitly restricted to analyzing thirteen predefined, high-impact business categories prevalent in Panabo City: Coffee Shops/Cafes, Print/Copy Centers, Laundry Shops, Car Washes, Food Kiosks/Stalls, Water Refilling Stations, Bakeries, Small Pharmacies, Barbershops/Salons, Motorcycle Repair Shops, Internet Cafes, Meat Shops, and Hardware/Construction Supplies. While the system evaluates singular enterprises, it also accommodates hybrid business models; however, this functionality is strictly capped at combining a maximum of two categories (a primary and a secondary business) per evaluation to maintain algorithmic stability.

For these specific sectors, the system employs dynamic spatial buffering... This evaluation relies on specific static data layers... and a newly integrated Hazard Risk Overlay. This overlay, visually represented by prominent red marks on the mapping interface, evaluates location vulnerabilities by cross-referencing proposed coordinates against localized hazard susceptibility maps. The presence of these red marks is directly bound to the spatial evaluation engine, actively penalizing the suitability score in heavy, moderate, and slight flood zones to ensure disaster resilience. Additionally, the scope now includes a Business Trend Suggestion page, which reverses the traditional search process by utilizing spatial data to proactively recommend highly viable business models to the user.

Despite these capabilities, the development and implementation of the study are bound by several operational and technical limitations. Due to current hardware and API constraints, the system cannot integrate live vehicle counts, real-time GPS foot-traffic flow, or live mobile tracking. Crucially, regarding the environmental risk assessments, MarketScope is not a real-time early warning system; the hazard evaluation relies entirely on static, historical shapefiles and data provided by the local government or relevant disaster management agencies, and it does not track live weather disturbances, active flood water levels, or real-time biological pest outbreaks. Furthermore, while the platform generates a spatial feasibility score, it is strictly not designed to perform operational budgeting, tax calculations, or deep accounting-level financial forecasting, such as return-on-investment or break-even analyses. The reliability and accuracy of the system's density and hazard maps also rely heavily on the frequency of manual data updates or the most recently available databases provided by the local government unit.

Additionally, the system does not track real-time individual consumer spending or private financial data. Instead, it utilizes spatial proxies such as building density, road network classifications, and proximity to complementary traffic generators and economic anchors to mathematically estimate market demand. The Automated PDF Reporting module utilizes a third-party Generative AI API (e.g., Google Gemini). To prevent AI hallucinations, the language model is strictly constrained to generate natural-language summaries based solely on the hard spatial, proxy, and historical hazard data queried by the system's backend, and it is neither designed nor legally authorized to provide licensed financial, real estate, environmental engineering, or legal counsel. The deployment of MarketScope is strictly restricted to a web-based application environment, optimized with a mobile-first responsive design to accommodate users accessing the system via smartphones. Finally, MarketScope does not guarantee business success; it will only give insights, comments, and suggestions.

Review of Related Literature and Works

This section explores the academic and technological foundations necessary for developing a geospatial decision support system tailored for small and medium enterprises. It details the transition from traditional, observation-based business placement methods to modern, data-driven site selection frameworks. By examining current academic methodologies in geographic information systems, proxy-based demand estimation, environmental resilience, and automated reporting, this review establishes the critical need for an accessible tool that mitigates financial risk for local entrepreneurs. Furthermore, it compares the proposed system against existing technological applications to contextualize its unique contributions to the local economic landscape.

Related Literature

Multi-Criteria Decision-Making (MCDM)

Historically, local entrepreneurs have relied heavily on observational heuristics when selecting business locations, a practice that significantly contributes to high failure rates. Modern research demonstrates a clear shift toward integrating Geographic Information Systems with multi-criteria decision-making models to address the remarkably complex problem of retail site selection. By synthesizing demographic and socio-economic variables to assess the quality of a potential location, utilizing fuzzy multi-criteria decision analysis allows business planners to mathematically evaluate the most optimal retail locations based on weighted empirical data rather than guesswork [7].

The Fuzzy TOPSIS Approach (Subjective & Linguistic)

A significant body of literature highlights the integration of Geographic Information Systems (GIS) with Multi-Criteria Decision-Making (MCDM) methods as a critical paradigm shift in transitioning from subjective, observation-based heuristics to empirical, data-driven site selection. A key study illustrating this development conducted a comparative analysis evaluating the optimal placement of shopping centers in Turkey by benchmarking two distinct methodologies: the expert-driven Fuzzy Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) and objective, data-driven GIS spatial layering. The findings demonstrated that while both independent models successfully converged on the highly commercial Marmara region as the primary target location, the subsequent rankings for the remaining regions fluctuated due to the respective trade-offs of each system namely, the subjective linguistic bias of Fuzzy TOPSIS and the rigid spatial determinism of GIS. This structural divergence underscores the risk of relying on isolated workflows and highlights the vital need for integrated frameworks that can simultaneously evaluate expert socioeconomic weights alongside hard spatial data layers. The architecture of MarketScope directly addresses this specific literature gap; by establishing a unified Intelligent Decision Support System (IDSS) that combines multi-criteria geoanalytics with local infrastructure proxies, municipal City Land Use Plan (CLUP) zoning compliance, and empirical environmental hazard layers, the system synthesizes the strengths of both data-driven spatial analysis and multi-criteria decision frameworks to optimize storefront placement for local MSMEs.

Fuzzy Real Options Valuation (FROV)

In data-limited environments where granular, street-level census records are unavailable, accurately estimating market demand requires innovative geospatial frameworks that substitute traditional demographic tracking with empirical infrastructure indicators. Researchers have successfully bypassed these local data constraints by utilizing volunteered geographic information (VGI) and open-source spatial datasets as reliable proxies for mapping micro-population distributions and commercial infrastructure density. Validated findings on open-source mapping confirm that localized building footprint density and road network classifications extracted from platforms like OpenStreetMap

(OSM) serve as highly effective proxy variables to quantitatively model human distribution, urban completeness, and localized market demand [9].

By extracting these spatial dependence features, modern systems can generate fine-resolution mapping without relying on expensive traditional census surveys. Beyond economic variables, an optimal site selection process must also account for environmental risks and municipal zoning. Research into urban resilience confirms that spatial decision support systems are critical for modeling natural hazards, demonstrating that evaluating proximity to flood accumulation zones and road networks significantly improves long-term business survival and disaster preparedness [10].

Related Works

Web-Based Point of Sale (POS) Systems

Web-based Point of Sale (POS) platforms represent the most common operational transaction architectures deployed by modern retail establishments to secure daily sales data [11]. These applications function primarily as local financial digitization solutions designed to streamline checkout processes and maintain real-time inventory adjustments for small businesses [12]. However, as architectural implementations show, standard POS frameworks operate entirely post-deployment, capturing business performance metrics only after a storefront has already been established 1[3]. Consequently, their backend data repositories are completely decoupled from geographic information networks, meaning they lack the analytical capacity to process spatial proximity parameters or evaluate competitor density arrays for pre-deployment site selection [14].

The Ideafy AI Platform and Cognitive Brainstorming Tools

The introduction of artificial intelligence within entrepreneurship education and startup incubation has spurred the development of conceptualization platforms like Ideafy AI [15]. These systems utilize natural language generation models to act as interactive co-pilots during the initial phases of corporate brainstorming and business model formulation [16]. While highly effective at optimizing text-based abstracts and evaluating linguistic business definitions, these platforms lack connection to physical geographic datasets [17]. Because they handle business data strictly as unstructured text input, they fail to integrate localized multi-criteria variables such as building footprint densities, environmental flood hazards, or micro-level infrastructure proxies [18].

Local Government Unit (LGU) Permit Trackers

Local Government Unit (LGU) permit trackers and e-permit applications serve as municipal e-governance solutions designed to handle document processing for new business registrations [19]. These systems are built upon standardized workflow routing models that guide an administrative application through multiple departmental validation checks [20]. While these e-governance systems effectively integrate municipal master plans and zoning boundaries for compliance screening, their scope is strictly regulatory [21]. They lack the analytical engines required to parse open-source road classification networks or compute automated suitability scores, leaving prospective entrepreneurs to identify physical market gaps without data-driven guidance [22].

Table 1. Comparative Table

Features	Web-Based POS	Ideafy AI Platform	LGU Permit Tracker	MarketScope
Geospatial Visualization			✔	✔
Automated PDF Reporting	✔	✔	✔	✔
Decision Support Score	✔	✔		✔
Zoning Integration			✔	✔
"Market Gap" Identification		✔		✔

As shown in table 1, MarketScope stands out as the only system among those reviewed that offers geospatial visualization, zoning integration, and market gap identification. While web-based POS systems, the Ideafy AI platform, and LGU permit trackers provide automated reporting or decision-support features suited for general business management or administrative processes, they are not designed to capture spatial proximity data for pre-deployment site selection. MarketScope fills these gaps by integrating automated feasibility reporting with multi-criteria geoanalytics tools tailored to the local economic ecosystem and zoning regulations of Panabo City.

Web-Based POS

Web-Based Point of Sale (POS) systems are standard transaction architectures used widely by retail businesses to manage daily sales and inventory. While Web-Based POS systems offer automated PDF reporting and digital financial records for operations, their design mirrors a general transaction model rather than a pre-deployment spatial analysis tool. As documented by Daniati et al., web-based POS platforms operate primarily as financial digitalization solutions without collecting or presenting geographic market data for site selection [23]. Web-Based POS systems also do not compute or display localized market gaps or competitor proximity, which is a core analytical feature of MarketScope for local entrepreneurs.

Ideafy AI Platform

The Ideafy AI Platform is an artificial intelligence-based application that assists entrepreneurs with startup analysis and conceptualization. While the Ideafy platform offers business ideation features and generates a decision support score based on text inputs, a review of AI-based startup platforms by Shah et al. found that these services focus primarily on generalized natural language generation and cognitive brainstorming, but lack localized spatial zoning integration and physical proximity metrics [24]. The Ideafy AI Platform is a general-purpose text generation platform not designed to provide environmental hazard intersections, zoning compliance checks, or physical market gap identification for defined urban ecosystems such as Panabo City.

LGU Permit Tracker

LGU Permit Trackers are localized e-governance platforms implemented by municipal governments to streamline the business registration process. These platforms operate on a workflow routing model, allowing local government units to manage documents and track application statuses in real-time. While this model offers effective zoning integration and administrative efficiency, it is not structured around competitive spatial analytics or proactive market research. According to studies on real-time business permit tracking systems, e-governance applications that focus on permit routing typically lack the infrastructure for collecting and presenting geospatial visualization, competitor density, or multi-criteria decision support scores calibrated for business viability [25]. As such, LGU Permit Trackers do not address the pre-deployment operational data gaps that MarketScope specifically targets for local small and medium enterprises.

Definition of Terms

Artificial Intelligence (AI) - The simulation of human intelligence processes by machines, particularly computer systems, which encompasses machine learning, natural language processing, and generative capabilities to perform complex cognitive tasks.

Building Density - An urban planning and spatial metric that measures the concentration of built structures within a specific geographic area, typically quantified by the ratio of total floor area or roof footprints to the land zone.

Business Trend Suggestions - A proactive spatial analytics function that evaluates unserved regional market voids, consumer demand signals, and geographic patterns to dynamically recommend highly viable commercial ventures for a targeted zone.

Catchment Radius - The defined geographic boundary or service area surrounding a commercial point of interest from which a business naturally attracts its primary consumer base and experiences immediate competitor friction.

City Land Use Plan (CLUP) - A comprehensive municipal policy document and regulatory framework that details the official structural vision and long-term legal zoning classifications for commercial, industrial, residential, and agricultural land allocations.

Flood Zone Layer - A localized environmental mapping dataset that defines areas susceptible to varying degrees of hydro-meteorological hazards, visually indicating geographic vulnerabilities bound to environmental evaluation criteria.

Geomarketing - A specialized branch of marketing and spatial economics that integrates geographic intelligence, location-based data, and mapping technologies to optimize commercial strategy, consumer demand modeling, and sales territory planning.

Geospatial Analysis - The comprehensive process of gathering, manipulating, displaying, and interpreting geographic data and spatial patterns to understand complex physical relationships across a defined landscape.

Hazard Risk Overlay - A conceptualized and digital layer of hazard susceptibility maps placed over base geographical layouts to evaluate, cross-reference, and expose physical vulnerabilities and structural exposure to natural disasters.

Hybrid Blending Logic - A multi-layered algorithmic computing routine designed to simultaneously calculate, merge, and output the combined mathematical feasibility and risk profile of two separate, co-located variables.

Intelligent Decision Support System (IDSS) - An advanced, interactive software architecture that integrates computational knowledge-bases, expert data models, and artificial intelligence to assist human decision-makers in solving poorly structured, complex problems.

Micro, Small, and Medium Enterprises (MSMEs) - Any business or commercial entity classified primarily by its total asset value and capitalization scales, functioning as essential drivers of local macroeconomic employment.

Multi-Criteria Decision Analysis (MCDA) - A structured, mathematical sub-discipline of operations research designed to evaluate, rank, and prioritize multiple conflicting criteria by applying objective weight values to complex decision problems.

OpenStreetMap (OSM) - A collaborative, open-source global mapping project designed to provide free, editable geographic data, points of interest, and infrastructure networks under an open-content license.

Spatial Proxies - Alternative, observable geographic data points utilized as mathematical substitutes to estimate unrecorded or private socio-economic variables, such as calculating regional purchasing power through local roof densities.

Suitability Score - A standardized numerical metric generated through quantitative modeling to classify the developmental or commercial viability of a targeted location based on weighted criteria.

Web-Based Mobile-Responsive Design - A flexible software interface development approach that enables a single application running on web-browser architectures to seamlessly scale, reposition, and optimize its user interface breakpoints across a full spectrum of desktop and mobile hardware screens.

CHAPTER 2
METHODOLOGY

This chapter presents the systematic process followed in the development of the MarketScope: Geospatial Analysis for Optimal Small-Business Placement with Integrated Reporting System. It describes the methods, procedures, and techniques used by the proponents to ensure that the system is developed in an organized, efficient, and reliable manner. The methodology serves as a guide in achieving the objectives of the study, particularly in creating a data-driven decision support system that assists entrepreneurs in selecting optimal business locations.

The model used in this research is the Hybrid Agile – Iterative Development Model. It enables the system to be developed through the repeated process of planning, designing, implementing, and evaluating the system in incremental cycles. This approach is highly effective for complex projects, as it allows for the rigid, upfront planning required for regulatory compliance while maintaining the flexibility of agile sprints needed for software engineering [26]. The approach is helpful in dividing the system into smaller components and enhancing their features individually. The iterative approach is also useful in identifying any issues and modifying the system according to the requirements.

It is ideal for the MarketScope framework, considering that it incorporates several complicated components, which include geospatial analytics, zoning regulations, hazard-risk mapping, and automated reporting based on AI. Furthermore, literature emphasizes that Web Geographic Information Systems (Web-GIS) exhibit characteristics of both standalone and web-based applications, making a hybrid methodology highly necessary for successful development and spatial data integration [27]. Each component of the framework undergoes a cycle of development and improvement where its effectiveness, accuracy, and usability are checked and optimized. The flexibility of the proposed approach will allow incorporating feedback into the process, improving efficiency. The system's development cycle consists of different key stages that include planning, requirement analysis, systems design, planning of modules or sprints, iterative development, integration, testing, validation, and review/iteration. This set of procedures helps implement the system and make sure it is functioning properly. In doing so, the developers make sure the MarketScope system delivers correct information when choosing the most appropriate place to open a business.

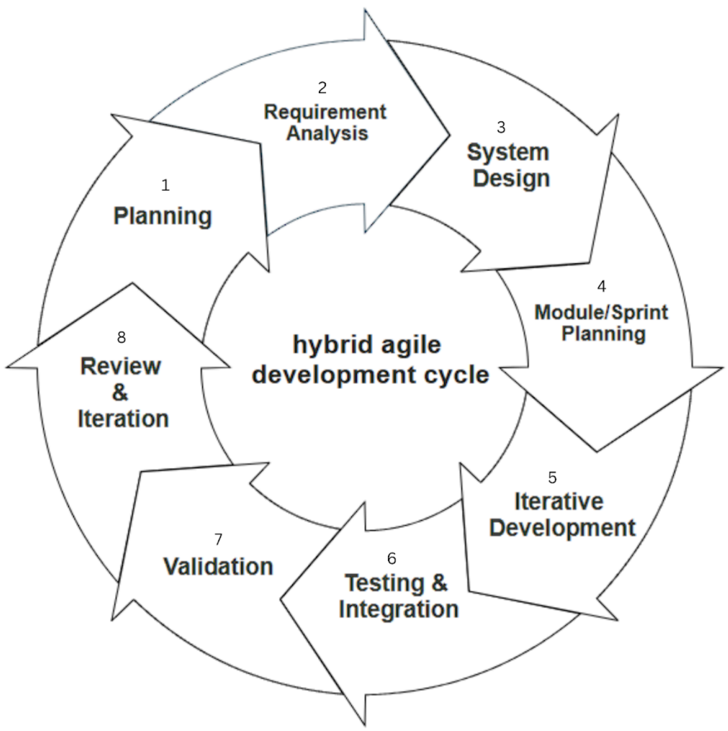

Figure 1. Process of Modified Hybrid Agile Development Model

System Planning

In the planning stage, the group identified the primary issue: poor location decisions for MSMEs in Panabo City due to the lack of data-oriented methods. They began to develop their project by establishing its boundaries, scope, as well as users, which were identified as business owners and entrepreneurs from the area of Panabo Proper. The team outlined their objectives, such as the creation of a visual representation of areas where businesses prevail, identification of gaps in the market, assessment of hazard risks, and development of feasibility reports based on artificial intelligence.

Requirements Analysis

In the Requirements Analysis stage, the requirements were obtained from both the functional and non-functional aspects based on the requirements identified in Chapter 1. The functional requirements were features such as visualizing the location of businesses in map format, analyzing market saturation and demand, integrating the City Land Use Plan (CLUP) zoning information, identifying the areas that are hazard-prone (e.g., flooding areas), generating suitability ratings, and generating feasibility reports generated by AI. On the other hand, the non-functional requirements pertained to making sure that the platform was mobile-responsive, had an intuitive interface, offered high-speed performance, and performed data processing efficiently. The proponents also listed down the critical data sources, which include OSM, CLUP zoning information, and hazard maps.

//Comparative Evaluation

- Identify the three model that you are going to use
- Discuss and cite why these three models that you have chosen for training
- Identify the metrics that you are going to use in finding the most accurate model
- Do comparative results in table format

System Design

The system architecture was planned and designed to allow for the integration of various parts which function together to achieve the desired functionalities of the MarketScope project. The front end will involve an interactive map as well as the user interface, whereas the back end will be responsible for computing and processing. The system makes use of a database to handle the spatial data, businesses, and zoning, and an artificial intelligence system to create feasibility reports. The modular design approach has been adopted to create the following modules, namely Geospatial Visualization, Spatial Evaluation, Zoning, Hazard Risk Overlay, and Artificial Intelligence Reporting.

Module / Sprint Planning

The system was divided into smaller, manageable modules through an Agile sprint approach. Every sprint emphasized on particular functionalities within the system to enable effective and well-organized development. Sprint 1 covered the map interface and geographic visualization features while Sprint 2 covered the business density and business competitor map functionalities. Suitability score analysis, which is based on a multi-criteria decision analysis engine, was covered in Sprint 3 while CLUP zoning functionality was covered in Sprint 4. Hazard risk analysis was covered in Sprint 5 with emphasis on flood zones, while report creation using AI technology was covered in Sprint 6.

Iterative Developement

In developing the system, an iterative process was adopted whereby improvements on the features were done at the end of every sprint. The iterative process included programming of system capabilities, incorporating data layering of space, scoring algorithm, as well as improving user interface design. Iterations made it possible for the developers to test and assess system performance. Comments gathered during the early phases of testing helped in improving system accuracy, efficiency, and functionality.

Testing and Integration

Each module underwent thorough testing before being integrated into the complete system to ensure functionality and compatibility. Various types of testing were conducted, including unit testing to verify individual components such as the scoring engine, integration testing to ensure that different modules work together seamlessly, system testing to evaluate the overall functionality of the system, and user testing to assess usability and performance. After successful testing, all modules including GIS mapping, AI reporting, zoning compliance, and hazard analysis were integrated into a unified system to ensure smooth operation and consistent output.

Validation

To guarantee the quality and relevancy of the results, the system was validated to ensure that it is in line with the aims of this research and produces credible results. System validation involved comparing the outputs from the system to the anticipated results, analyzing the accuracy of the suitability ratings produced by the system, analyzing the output generated for zoning compliance, and analyzing the relevance of the reports generated using the AI technology.

Review & Iteration

The system was continually reviewed and refined following its validation stage in order to improve its efficiency and effectiveness even further. This involved identifying various problems and shortcomings that existed with the system, improving the accuracy of the algorithms used in the system, improving usability of the user interface, and generally enhancing the performance of the system.

Project Team Organization

The roles and responsibilities of the project development of the system will be explicitly stated and are essential to the effective execution of the project. A systematic process will take place wherein the work will be allocated appropriately among the team members and performed efficiently.

Figure 2 shall display the organizational chart for MarketScope: Geospatial Analysis for Optimal Small-Business Placement with Integrated Reporting System. The overall responsibility for the completion of the whole development process rests upon the project manager who will coordinate all activities related to the system's design and development in compliance with the project goals and end-user requirements.

Figure 2. The Project Development Team Chart

The system developer is assigned the tasks of designing, developing, and implementing the system, including both its front end and back end components. He/She shall integrate the geographical information systems, zone data and hazard layers in the software. Further, he/she will be required to design the spatial evaluation engine and the artificial intelligence reporting module. The developer shall make sure that all features of the system including mapping, analysis, scoring, and reporting work well together.

The documentarian will take charge of collecting, organizing and keeping all project related documentation. The documentarian will prepare all necessary technical documents, system documentation and user manuals.

Work Breakdown Structure

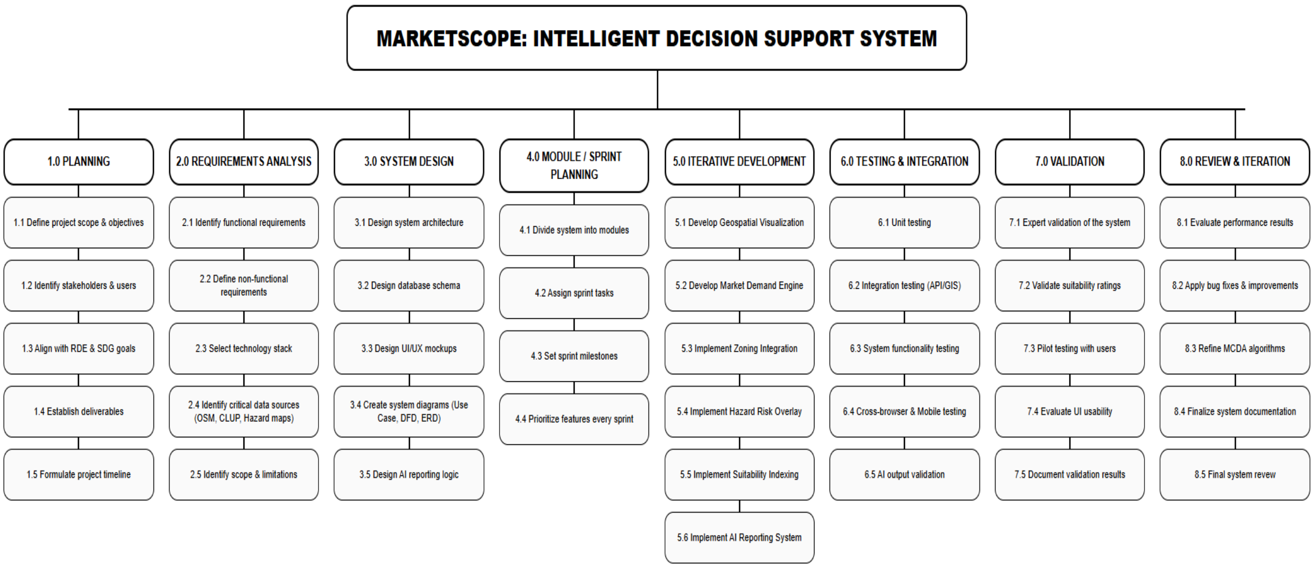

Figure 3. Work Breakdown Structure of MarketScope

Figure 3 demonstrates the Work Breakdown Structure (WBS) of the MarketScope project, depicting the process of development in a hierarchical and structured form. The entire project will be divided into sub-processes and phases in order to develop and manage all of the system elements efficiently. Using the WBS tool to structure the development of the project will allow for visibility of the entire development process and will ensure efficient planning and time management throughout each sprint phase. Some of the main phases in which the development of the project will be undertaken include Geospatial Mapping, Business Density Analysis, Zoning Integration, Hazard Risk Assessment, Suitability Scoring, and Report Generation through the use of Artificial Intelligence. When developing the MarketScope application, using the WBS approach will allow to undertake all the development processes efficiently and logically, allowing implementation of the Agile Iterative Development model.

Gantt Chart

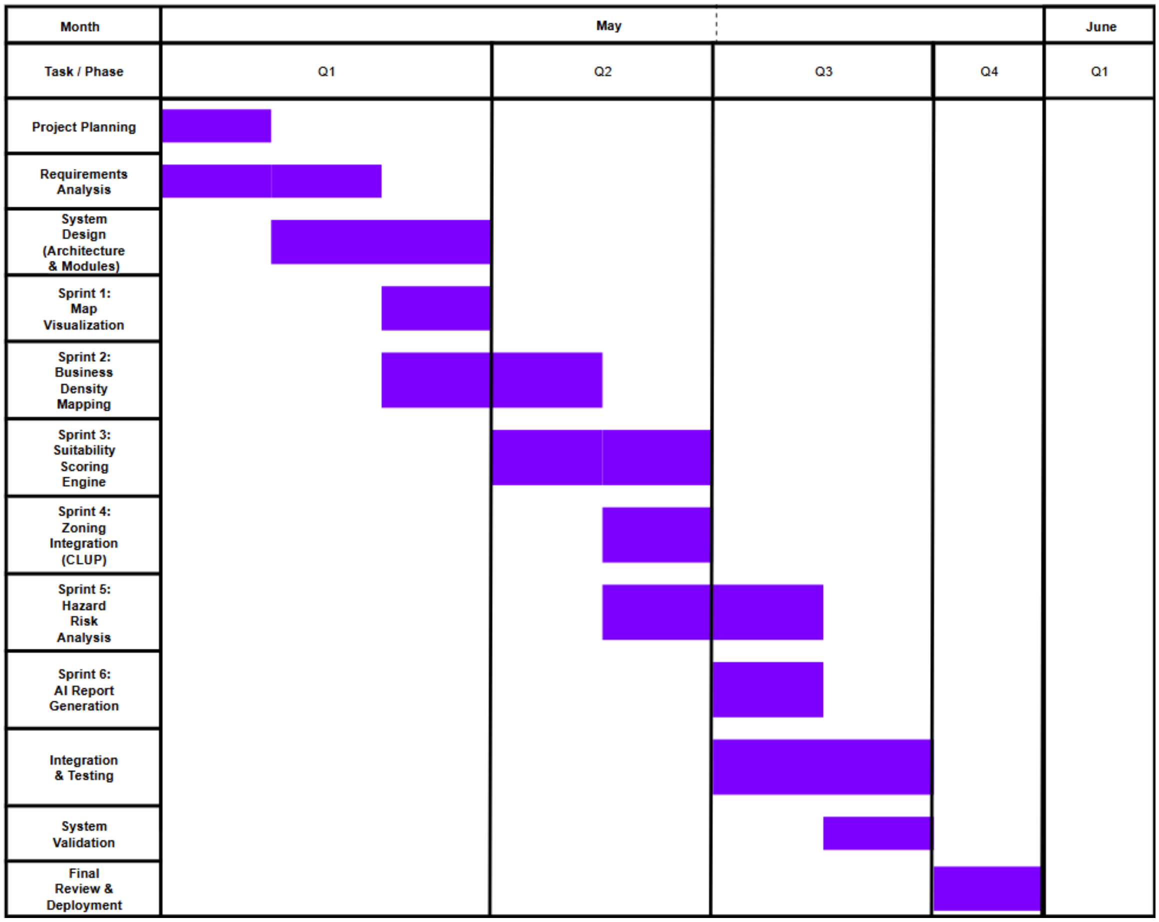

Figure 4. Gantt Chart of the Project of MarketScope

Figure 4 outlines a structured, hybrid project timeline for a specialized software development lifecycle, likely a geographic information system (GIS) or urban planning tool spanning from the first quarter to the fourth quarter. The project begins in the first quarter with foundational Project Planning and Requirements Analysis, quickly transitioning into a lengthy System Design phase that overlaps with the earliest development increments. As shown in Figure 4, development is organized into six functional milestones, starting with foundational core components like Map Visualization late in the first quarter and progressing through analytical modules like Hazard Risk Analysis and AI Report Generation by the middle of the third quarter. The timeline heavily utilizes parallel scheduling, particularly in the second and third quarters, where multiple development sprints run concurrently alongside a comprehensive Integration & Testing phase that spans the entirety of the third quarter. The project concludes with a dedicated period for System Validation at the tail end of the third quarter, leaving the initial block of the fourth quarter exclusively for Final Review & Deployment, which ensures a robust buffer for final quality assurance before launch.

System Analysis

System analysis is concerned with defining the problems at hand and specifying the requirements needed in developing the MarketScope system. According to Chapter 1 findings, most MSMEs and entrepreneurs use their intuition ("gut feel") in determining business location, leading to bad choices, market overcrowding, and higher chances of failure. Therefore, there exists a need for a system that will use data to enable better decision making in selecting the best sites for businesses. This step involves making sure that the proposed solution addresses all these problems through systematic analysis.

From our study of the current scenario, it is quite evident that there is no integrated solution to address the geospatial and zoning data as well as hazardous area issues within Panabo City. Current business entrepreneurs do not have access to systems that can help in evaluating the demand for markets and businesses and determine the level of competition within those areas and even environmental hazards, such as floods and earthquakes. Thus, there is a dire need for an intelligent system capable of integrating complicated datasets into useful decision support mechanisms.

The suggested system of MarketScope will fill these gaps by combining the use of geospatial analysis, multiple criteria decision-making, and AI reporting in one application. The system takes into account several aspects such as the business density of the area, accessibility of the infrastructure, zoning, and hazards to develop the suitability score of the selected site. Moreover, MarketScope allows users to analyze the results on the maps and obtain a feasibility report automatically. With its help, it is possible to convert the spatial data into relevant information to ensure strategic decisions regarding business placement.

In general, this system analysis proves the necessity of developing an intelligent system to facilitate the process of making decisions in favor of MSMEs. The analysis has outlined the shortcomings of the current approaches and the benefits of integrating technology into the business planning process.

System Architecture

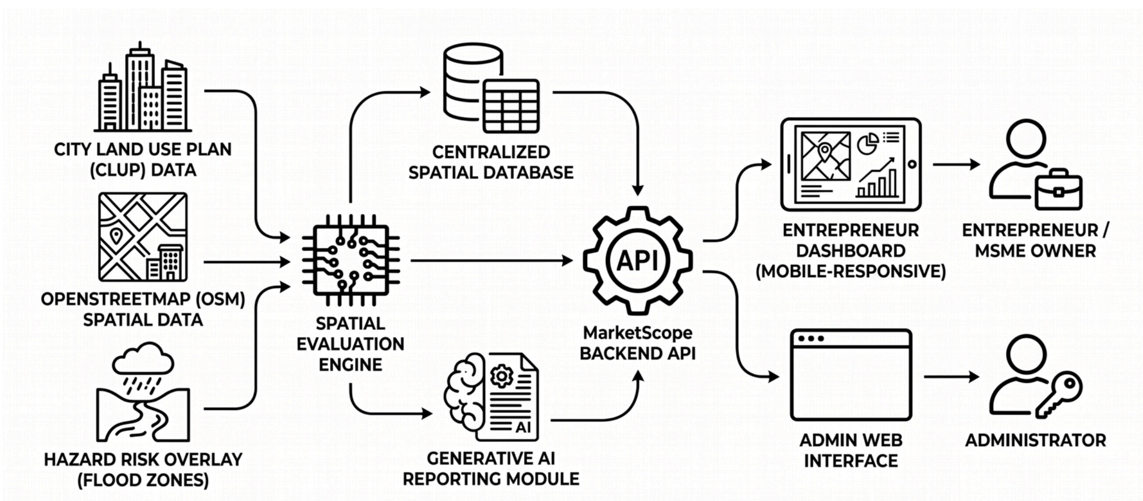

Figure 5. System Architecture of MarketScope

The system architecture is composed of several interrelated layers that are necessary to conduct geospatial analysis, data processing, and automated reporting. Data Layer contains information collected by the system, such as existing business locations, road networks, infrastructure data, CLUP zoning, and hazard risk maps. Processing Layer is responsible for conducting the analysis of collected information via the use of geospatial analysis, market demand, and competition, conducting zoning analysis, evaluating hazard risks, and calculating a suitability score. The information is then managed by Backend Layer via connecting the database, spatial evaluation engine, and AI reporting module. Application Layer contains the web-based, responsive interface with the following options: selection of a business location, selecting a business category, map results, zoning/hazard evaluation, and generating feasibility reports. The access to the application is controlled using user authentication, allowing administrators to manage data, system records, and reports and regular users to analyze locations and export results.

Conceptual Framework

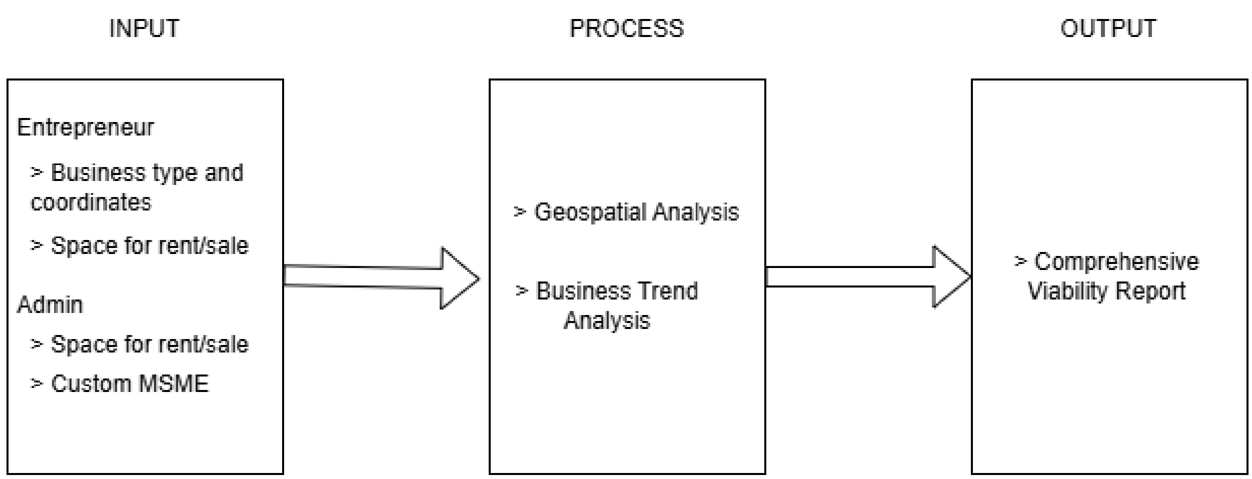

Figure 6. Conceptual Framework

Figure 6 illustrates the conceptual framework of MarketScope: Geospatial Analysis for Optimal Small Business Placement with Integrated Reporting

System, a web-based Decision Support System meant to aid MSMEs in choosing the best location for setting up their businesses, is shown in Figure 6. The input layer has spatial data like business locations, road network data, OpenStreetMap data, hazard risk information (like floods), and various parameters selected by the user. In the process layer, the technology that is used is geospatial analysis, decision making engine, and intelligent report generation using AI. The processing of data uses a web-based system along with the mapping technologies, database which stores all the spatial and business data, and AI module used for creating the feasibility report.

Functional Requirements

1. Visualizes business locations, infrastructure, and hazard-prone areas using a web-based mapping interface.
2. Analyzes market demand and competition based on spatial data such as building density and road networks.
3. Calculates a suitability score for selected locations using a multi-criteria decision-making approach.
4. Integrates City Land Use Plan (CLUP) data to ensure zoning compliance.
5. Identifies underserved market areas and potential business opportunities.
6. Evaluates environmental risks such as flood-prone zones.
7. Generates automated feasibility reports using an AI-powered reporting module.
8. Allows users to analyze and compare up to two business types through hybrid evaluation.
9. Stores spatial data, business data, and analysis results in a centralized database.
10. Provides a mobile-responsive web interface for accessibility across devices.
11. Assesses environmental hazards, such as floodplains, and displays them as red map overlays that automatically adjust the overall suitability score.

12. Examines spatial dynamics and market conditions to preemptively identify viable business categories, streamlining the venture discovery process.
13.

Non- Functional Requirement

1. The system must display maps, analysis results, and reports through a web-based dashboard for easy access.
2. The system must be accessible through modern web browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.
3. The system must be compatible with desktop, tablet, and mobile devices.
4. The system interface must be simple, clear, and user-friendly for entrepreneurs.
5. The system must ensure accurate and reliable analysis of spatial and business data.
6. The system must maintain stable performance when processing multiple user requests.
7. The system must ensure data security and protect stored information from unauthorized access.
8. The system must provide fast processing of user queries and analysis results.

Use Case Diagram

The Use Case Diagram for the MarketScope system illustrates how users interact with the platform to analyze business locations and generate feasibility reports. The primary user, the entrepreneur or MSME owner, interacts with the system by selecting a location, choosing a business category, and viewing analysis results. The system processes user inputs by performing geospatial analysis, evaluating market demand and competition, checking zoning compliance, and assessing hazard risks. The AI module then generates a feasibility report based on the processed data. The system allows users to view maps, analyze suitability scores, and download reports, ensuring a complete and user-friendly decision support process.

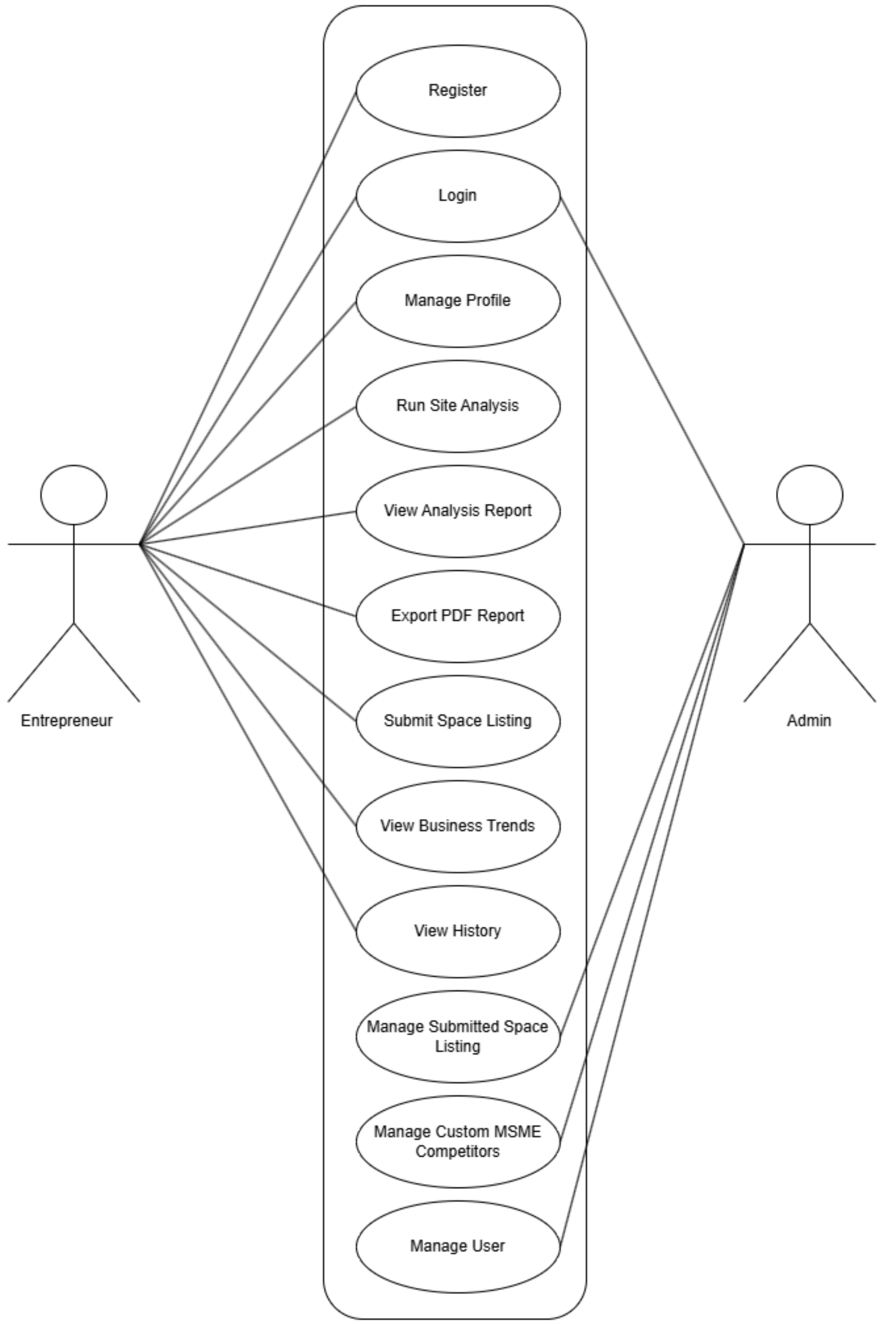

Figure 7. Use Case Diagram of MarketScope

Context Flow Diagram

The context flow diagram depicts the MarketScope system as an integrated system responsible for managing information interactions between the system and its data sources. The inputs to the system include the chosen parameters by the user like the choice of the location, business, and analytical needs, but the system processes the spatial data which includes the location of business units, infrastructure, zoning data, and hazards. However, the output produced by the system includes suitability maps, market gap analysis, zoning analysis, and automatic feasibility report generated through artificial intelligence. Here the user makes the request of analysis which is processed by the system, and in return, the user gets information.

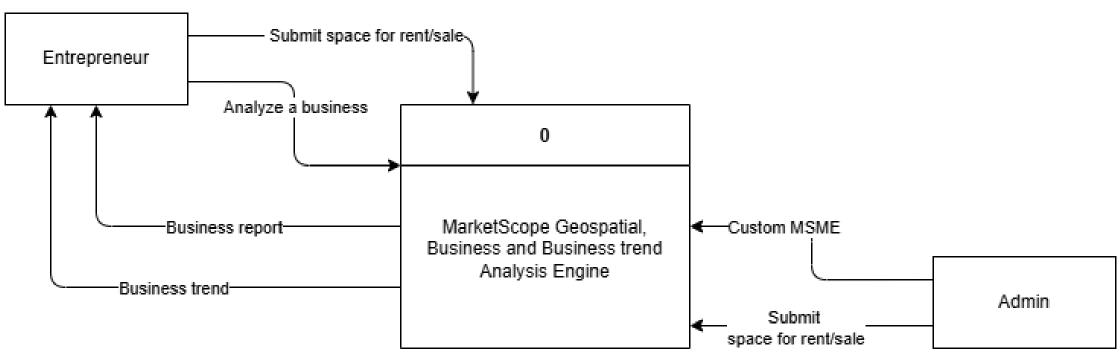

Figure 8. Context Flow Diagram of MarketScope

Data Flow Diagram

The Data Flow Diagram represents the primary processes in the MarketScope application as well as illustrates data flow across all platforms of the application. Input from the users will include parameters like selected location, chosen business sector, and user-defined options for analysis. This input will be fed into various modules of the system, including geospatial mapping, analysis of market demand and competition, zoning analysis, hazard risk analysis, generation of suitable scores, and AI report production. Database data to be accessed by the system will include business locations, infrastructural data, CLUP zoning data, and hazard data. Once data is processed by the application, output will be provided to the user including visualization, suitable scores, market gap analysis, zoning status, hazard risk assessment, and download of feasibility reports.

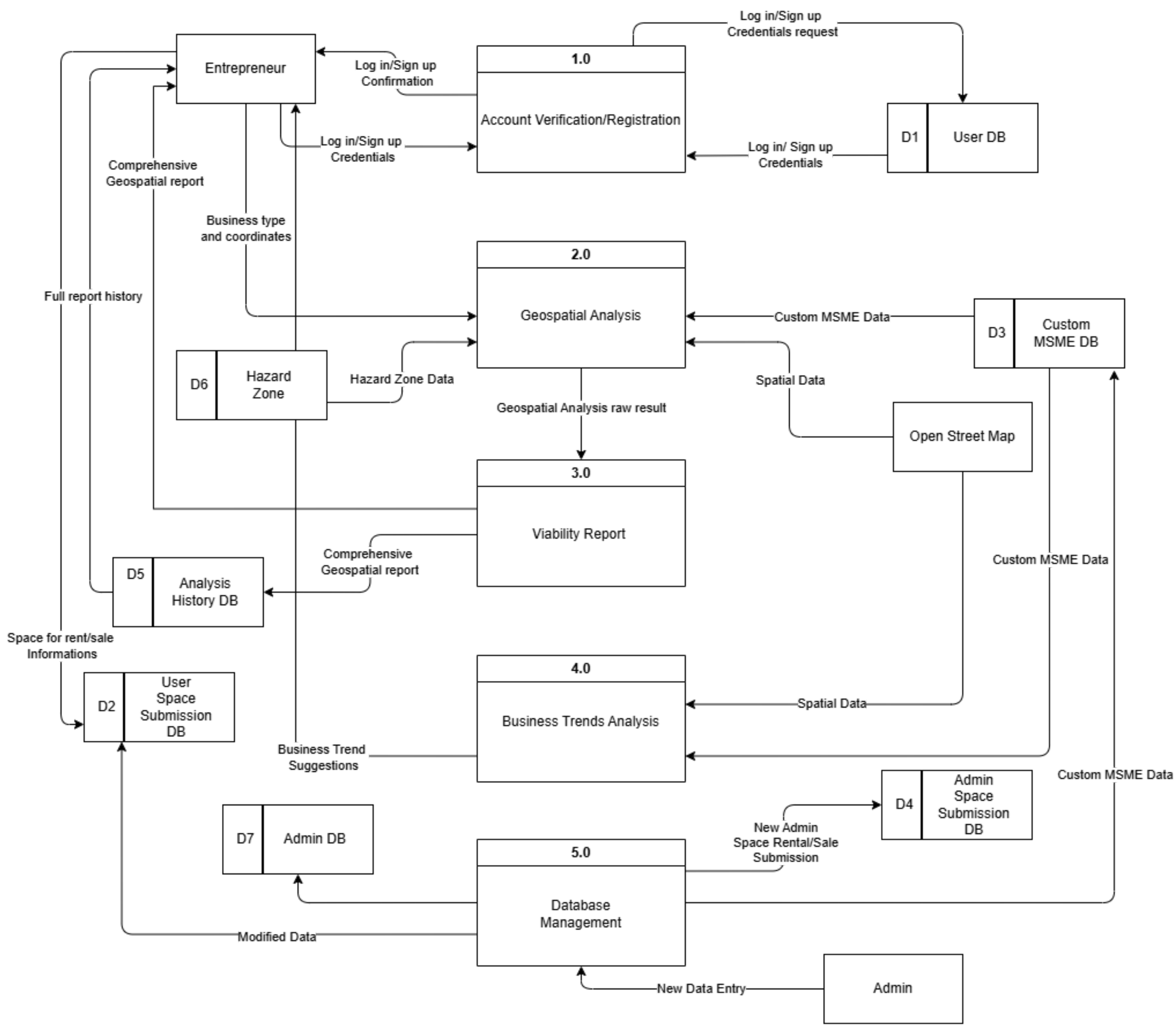

Figure 9. Data Flow Diagram of MarketScope

Entity Relationship Diagram

The Entity Relationship Diagram (ERD) provides the database architecture of the MarketScope system and highlights the connection between the key data entities. Some of the crucial data stored in the database include user information, selected business categories, business location, zoning data, hazard risk zones, suitability score, and the feasibility report generated. The user entity will consist of the account information and will be related to the analysis records generated by the user. In turn, each analysis record will include the selected business location, selected business category, zoning assessment, hazard risk assessment, and overall suitability score. On the other hand, the business category entity will indicate the various MSME categories that can be analyzed using the platform. Finally, the spatial data entity will include data such as the existing businesses, infrastructure data, CLUP zoning data, and flood-prone zones.

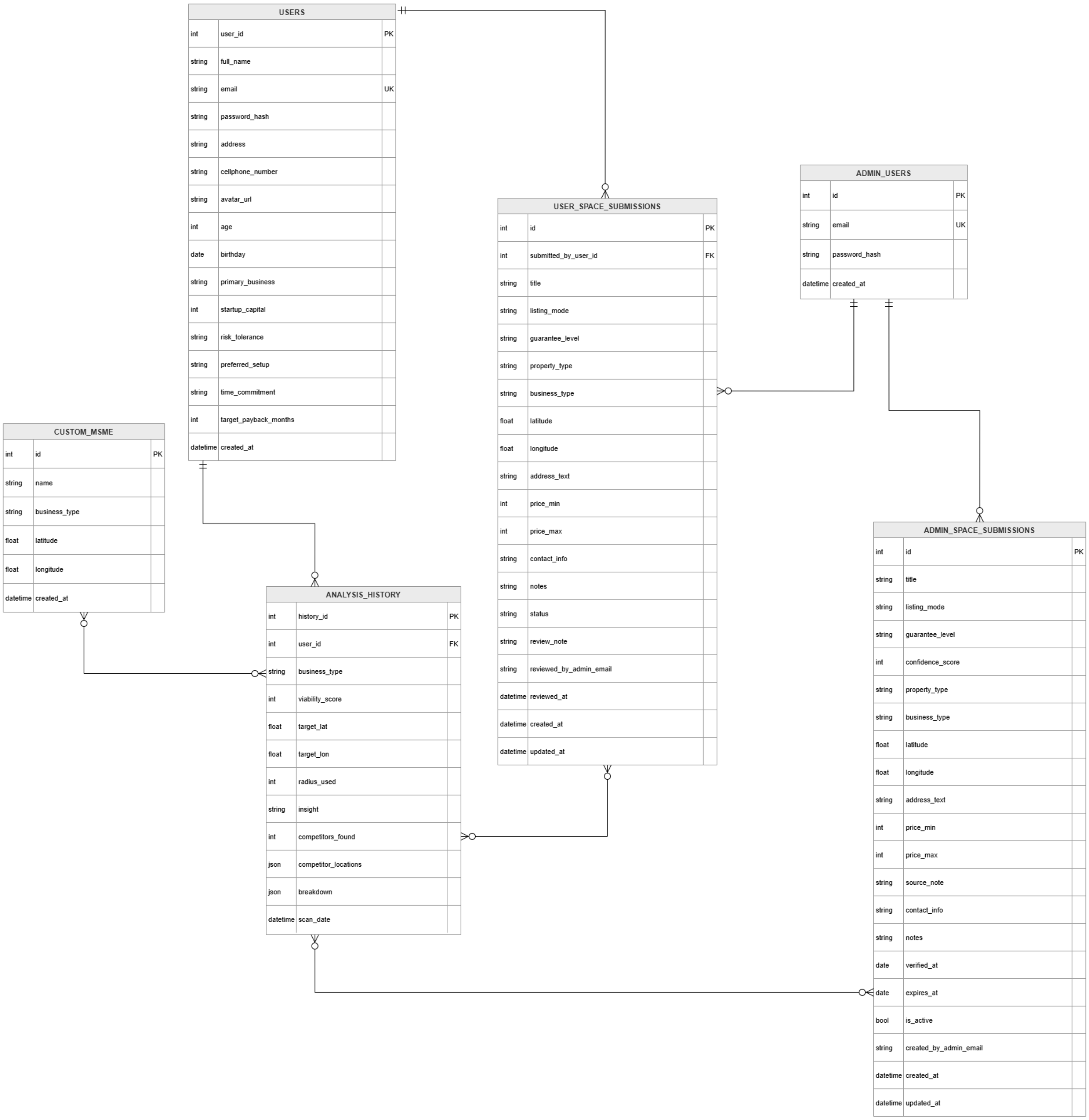

Figure 10. Entity Relationship Diagram of MarketScope

Data Dictionary

The Data Dictionary defines the structure, type, and purpose of all data used in the MarketScope system, ensuring consistency and accurate data management for analysis and reporting.

Table 2. Data Dictionary

USERS

Column name	Datatype	Constraint	Description
user_id	Integer	Primary Key	Unique identifier for the user.
full_name	Varchar(255)	Not Null	The complete name of the entrepreneur.
email	Varchar(255)	Unique, Not Null	Email address used for authentication.
password_hash	Text	Not Null	Encrypted (bcrypt) password string.
address	Text	Not Null	Physical address of the user.
cellphone_number	Varchar(32)	Nullable	Contact number of the user.
primary_business	Varchar(120)	Nullable	The user's main industry of interest.
startup_capital	Integer	Nullable	The estimated budget of the user for business planning.
risk_tolerance	Varchar(20)	Nullable	User's risk preference for system recommendations.
created_at	Timestamp	Default: CURRENT	Date and time the account was created.

Admin User

Column name	Datatype	Constraint	Description
id	Integer	Primary Key	Unique identifier for the admin.
email	Varchar(255)	Unique, Not Null	Admin email address for login.
password_hash	Text	Not Null	Encrypted password string.
created_at	Timestamp	Default: CURRENT	Date and time the admin account was created.

Analysis History

Column name	Datatype	Constraint	Description
history_id	Integer	Primary Key	Unique identifier for the user.
user_id	Varchar(255)	Not Null	The complete name of the entrepreneur.
business_type	Varchar(255)	Unique, Not Null	Email address used for authentication.
viability_score	Text	Not Null	Encrypted (bcrypt) password string.
target_lat	Text	Not Null	Physical address of the user.
target_lon	Varchar(32)	Nullable	Contact number of the user.
radius_used	Varchar(120)	Nullable	The user's main industry of interest.
insight	Integer	Nullable	The estimated budget of the user for business planning.
competitors_found	Varchar(20)	Nullable	User's risk preference for system recommendations.
competitor_locati	Timestamp	Default: CURRENT	Date and time the account was created.

ons			
breakdown	JSONB	Nullable	Detailed JSON object storing sub-scores (zoning, hazard).
scan_date	JSONB	Default: CURRENT	When the analysis was executed.

User Space Submission

Column name	Datatype	Constraint	Description
id	Integer	Foreign Key	Unique identifier for the listing.
submitted_by_user_id	Integer	Foreign Key	Links the listing to the submitting `users(user_id)`.
title	Varchar(255)	Not Null	The display name of the space listing.
listing_mode	Varchar(12)	Not Null	Identifies if the space is for Rent or Sale.
property_type	Varchar(80)	Nullable	Defines if it is an empty lot, commercial space, etc.
latitude	Double Precision	Not Null	Latitude coordinate of the available space.
longitude	Double Precision	Not Null	Longitude coordinate of the available space.
price_min	Integer	Nullable	Minimum price range in PHP.
status	Varchar(20)	Default: CURRENT	Workflow state (pending, approved, rejected).
reviewed_by_admin_email	Varchar(255)	Nullable	Admin who processed the submission.

Technologies, Concepts, and Theories

The MarketScope system combines the principles of geospatial analytics and artificial intelligence to provide data-driven decision-making in identifying the best possible sites for small businesses. Geographic data such as the geographic positions of various businesses, roadways, infrastructural assets, zoning details, and risk factors for disasters are gathered using OpenStreetMap data and local government data sets. The data are then analyzed using a web-based application that employs mapping techniques, a central database, and an artificial intelligence-based report generator.

Geospatial analysis and Multi-Criteria Decision Analysis (MCDA) are extensively utilized in location-oriented decision support systems that consider various criteria including demand, competition, accessibility, and environmental hazards. Geo-analysis and MCDA are employed in MarketScope to compute the suitability score through spatial association and indirect measurements such as building density and adjacency to critical infrastructures. Through this approach, MarketScope can uncover overlooked regions, determine market voids, and offer sound guidance on business establishment.

The system is also based on three theoretical concepts. The concept of General Systems Theory promotes the integration of spatial data, modules of data processing, and interaction with users to form one decision support system. Data-driven decision-making theory emphasizes the importance of using data analyzed and reported to facilitate entrepreneurship and proper decision-making. The concept of sustainable development theory assists the above theories in balancing urban development to ensure that businesses are placed according to zoning laws and policies.

System Testing and Implementation

System testing and implementation stage of the MarketScope system guarantees the successful performance of the application and its readiness for practical use. Testing involves verification of the application's ability to perform all its tasks efficiently including geospatial analysis of market demand and competition, zone checking, hazard evaluation, and calculating of suitability scores and providing feasibility reports. In addition, testing will ensure that map visualization is correct and the output of AI reporting is useful and accurate.

Implementation involves well-organized steps, starting with setting up the web platform and its functionality, followed by integration of geospatial layers into the system including business locations, zones, hazards, and other data, and availability of the system to users through web browsers. Users are instructed on how to operate the system correctly, while data is stored and maintained in a secure manner.

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System Maintenance

System maintenance will be performed to ensure the long-term reliability, accuracy, and usability of MarketScope. This will include updating business location records, infrastructure data, CLUP zoning information, and hazard risk maps whenever new or updated datasets become available. Maintenance will also involve fixing errors, improving system performance, enhancing security, and refining the user interface based on feedback. Since the system relies on spatial and local government data, regular updates will be necessary to maintain the accuracy of the analysis results. Continuous maintenance will help ensure that MarketScope remains useful, relevant, and effective as a decision support tool for MSMEs and entrepreneurs in Panabo City.

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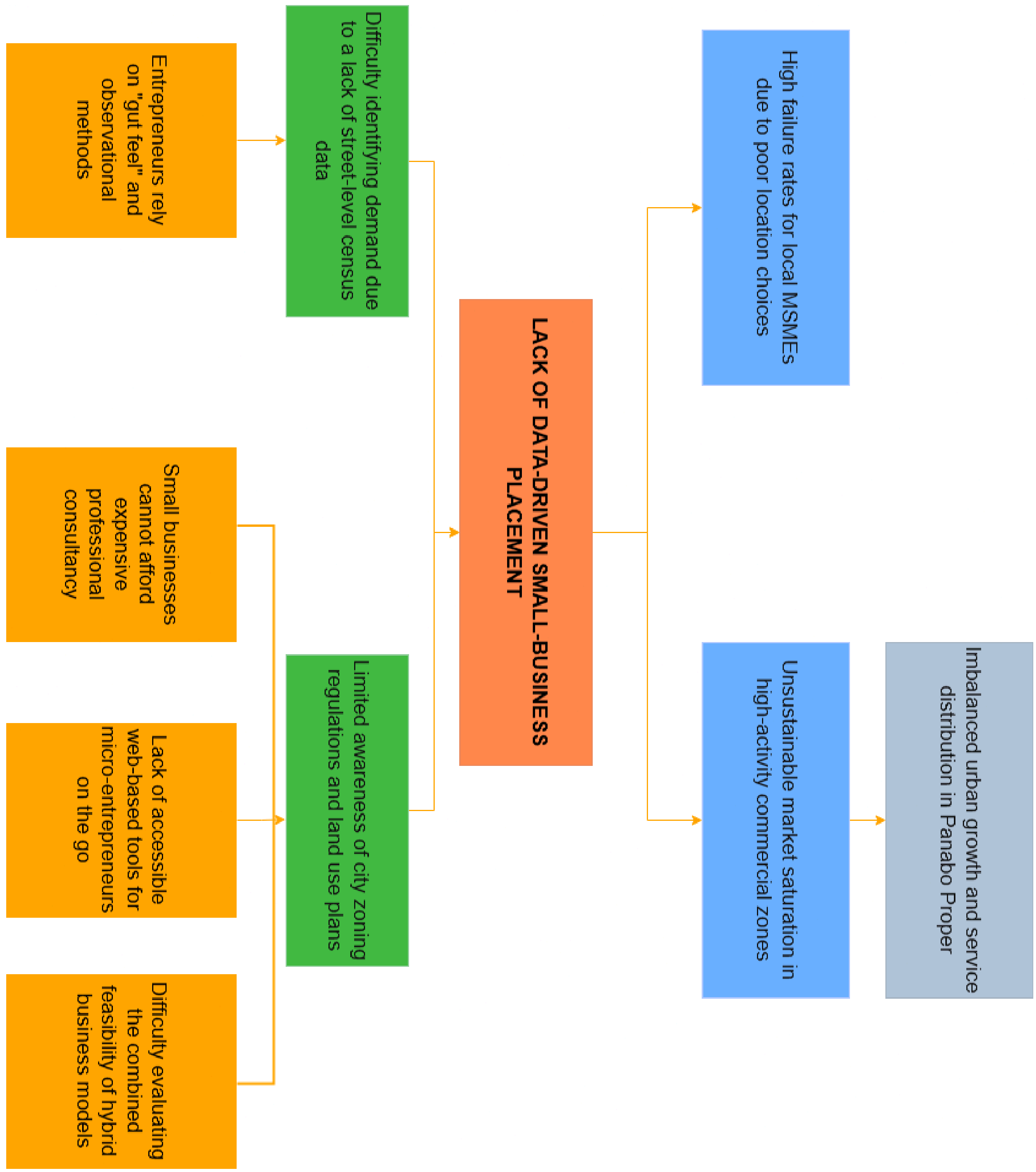

Figure 1. Problem Tree Analysis Diagram

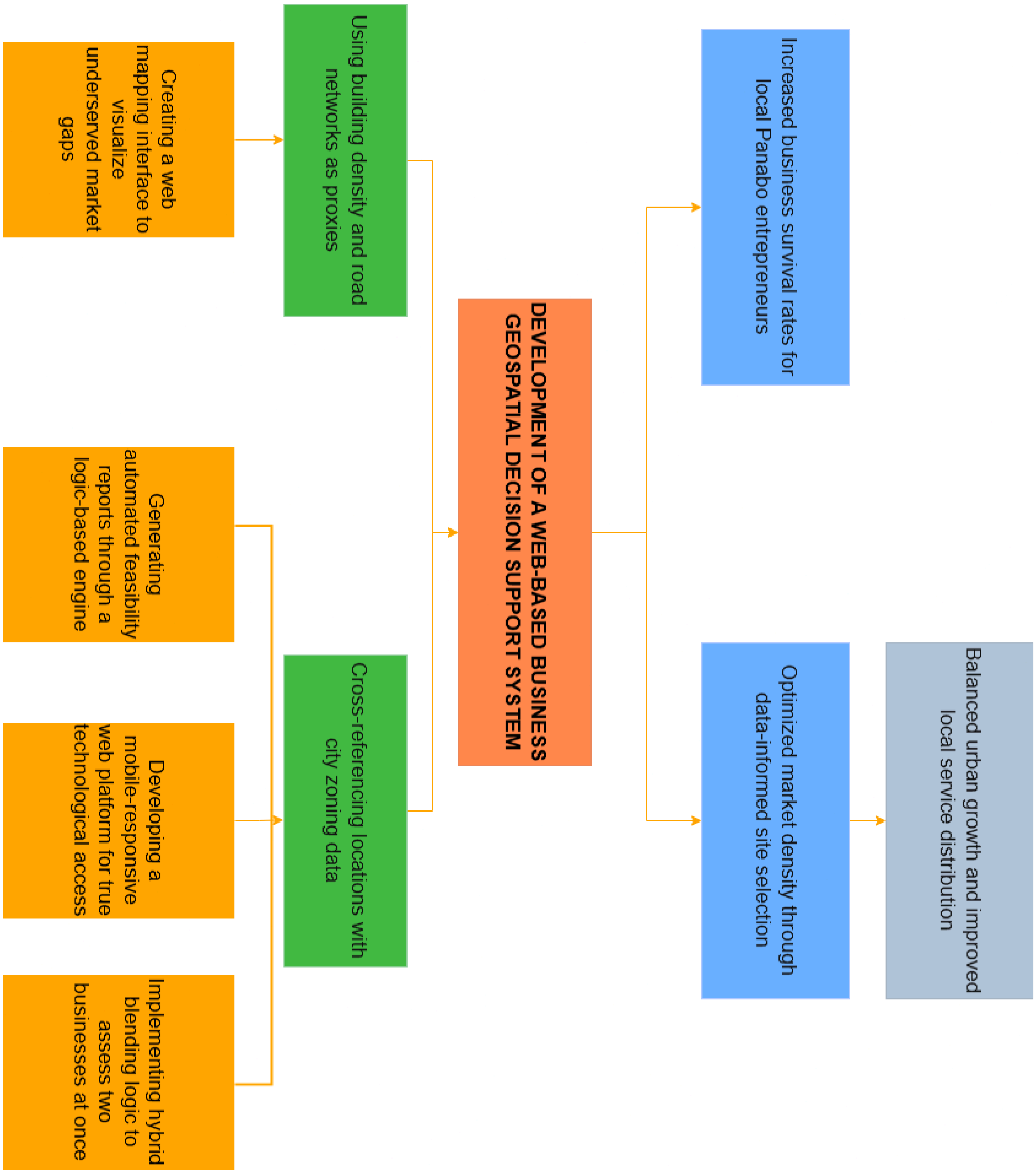

Figure 2. Objective Tree Analysis Diagram

Features	**Web-Based POS**	**Ideafy AI Platform**	**LGU Permit Tracker**	**MarketScope**
Geospatial Visualization			✔	✔
Automated PDF Reporting	✔	✔	✔	✔
Decision Support Score	✔	✔		✔
Zoning Integration			✔	✔
"Market Gap" Identification		✔		✔

Table 1. Comparative Table